

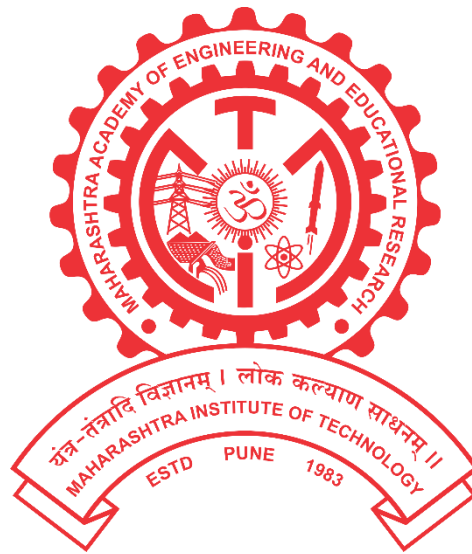
A Seminar on
Variable Compression Ratio Engines

By

Mr. Gaurav Santosh Gokhale

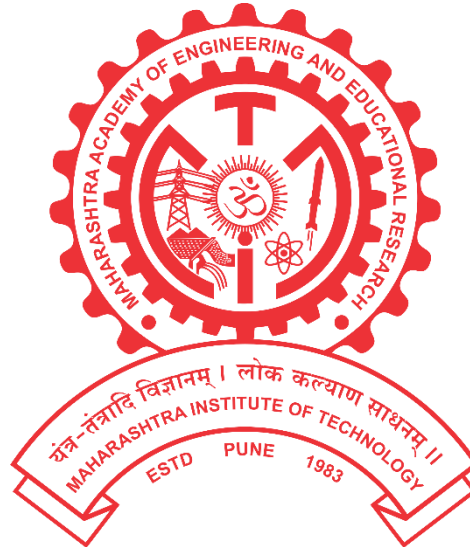
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C E R T I F I C A T E

This is to certify that Mr. Gaurav Santosh Gokhale, Seat No T150020837 has successfully completed the Seminar work entitled “Variable Compression Ratio Engines” under my supervision, in the partial fulfilment of Bachelor of Engineering - Mechanical Engineering, by Savitribai Phule Pune University.

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Gaurav Gokhale

Abstract

Increasingly stringent emissions and fuel economy standards have long remained a source of challenges for research in automobile engine technology development towards the more thermally efficient and less polluting engine. Variable compression ratio (VCR) technology has long been recognized as a method for improving the fuel economy of SI engines. Spark ignition (SI) engines have lower part-load efficiency when compared with the diesel engines. To improve this efficiency operate SI engine with high compression ratio at part load and with low compression ratio at high load. Recent research at Nissan on a variable compression ratio (VCR) mechanism that uses multiple links to form a new piston/crank system has revealed new potentialities for the multiple-link mechanism in addition to its intended VCR capability. By continuously changing the compression ratio, an engine enables optimum combustion efficiency at all engine speed and load conditions, realizing better engine performance, lower fuel consumption, and lower exhaust emissions. The compression ratio can range from high (14:1 maximum) bringing high efficiency to low (8:1 minimum) delivering high performance and a strong synergy with turbocharging; it can be said that this engine can intelligently and autonomously change its characteristics based on the driver's demand.

Contents

1. Introduction
 - 1.1 Various ways of implementing VCR
2. What is Compression ratio?
3. Effects of Compression Ratio
 - 3.1 Compression ratio and thermal efficiency
 - 3.2 Compression ratio and power
 - 3.3 Turbocharger and compression ratio
 - 3.4 Compression ratio and brake specific fuel consumption
4. What is Variable Compression Ratio Engine
 - 4.1 Overview
 - 4.2 Difficulty in operation
5. Case Study: INFINITI-Nissan VCR-Turbo Engine
 - 5.1 Overview
 - 5.2 Construction Of VCR-Turbo
 - 5.3 Operation
 - 5.4 Salient Features
 - 5.4.1 Reduction in side thrust during combustion stroke
 - 5.4.2 Decrease in BSFC
 - 5.4.3 Contribution to output and torque improvement
6. Summary
7. References

List of figures:

- Figure 1.-Compression ratio-Page 7
- Figure 2.-Compression Ratio and Thermal Efficiency-Page 8
- Figure 3.-Maximum Reliable boost pressure graph-Page 11
- Figure 4.- Engine Load & BSFC Page 12
- Figure 5.-Strategies to increase fuel efficiency-Page 13
- Figure 6.-Equivalent Line diagram of multilink mechanism-Page 15
- Figure 7.-Construction of VCR-Turbo-Page 18
- Figure 8.- Operation of VCR-Turbo-Page 19
- Figure 9.-Reduction in side thrust-Page 20
- Figure 10.- Decrease in BSFC-Page 21
- Figure 11.-Engine speed Vs Engine Torque (VCR-Turbo)-Page 22

Chapter 1

INTRODUCTION

Variable Compression Ratio is a technology to adjust the compression ratio of an internal combustion engine while the engine is in operation. This is done to increase fuel efficiency while under varying loads. Higher loads require lower ratios to be more efficient and vice versa. Variable compression engines allow for the volume above the piston at the ‘Top dead center’ to be changed. For automotive use this needs to be done dynamically in response to the load and driving demands. Spark ignition engines have a limit on the maximum pressure during the compression stroke. By adopting variable compression ratio best of both worlds can be achieved. This is done by changing the size of combustion chamber.

1.1 Various ways of implementing VCR [5]

- Articulated cylinder head
- Hydraulic Pistons
- Eccentrics on Bearings
- Multilink rod-crank mechanisms
- Additional piston in cylinder head
- Gear based mechanisms

In this paper the case of Infiniti-Nissan VCR-Turbo engine is discussed which contains the Multilink rod-crank mechanism.

Chapter 2

What is compression ratio?

The static compression ratio of an internal combustion engine is a value that represents the ratio of the volume of its combustion chamber from its largest capacity to its smallest capacity. It is a fundamental specification for many common combustion engines. In a piston engine, it is the ratio between the volume of the cylinder and the combustion chamber when the piston is at the top of its stroke.

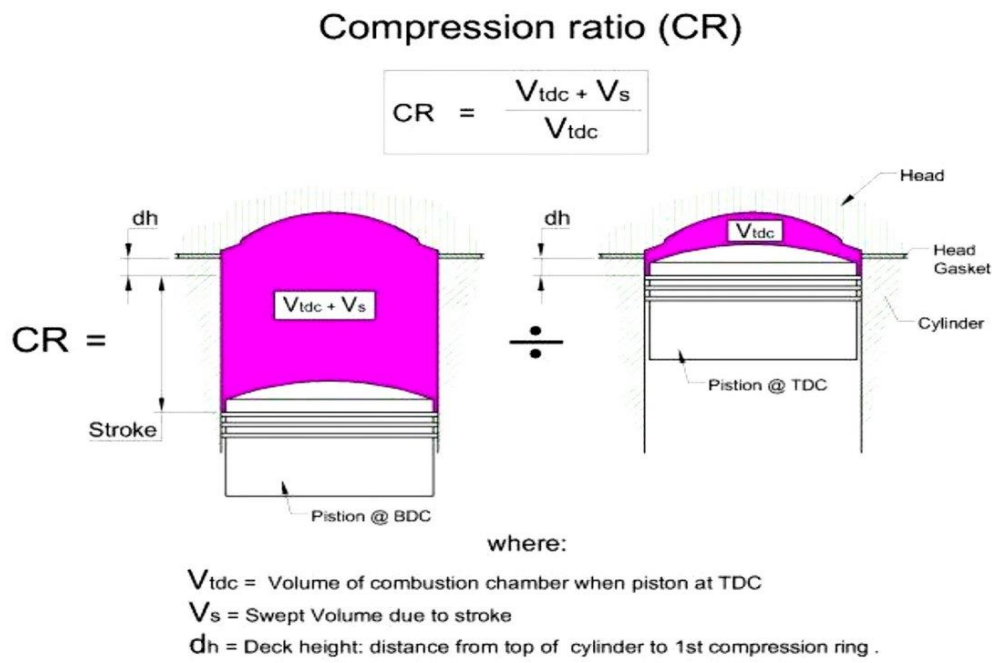


Figure 1.-Compression ratio

The value of compression ratio for passenger cars ranges from 8:1 to 10:1 while compression ratio for diesel engine ranges from 12:1 to 23:1

Chapter 3

Effects of Compression Ratio

3.1 Compression ratio and thermal efficiency

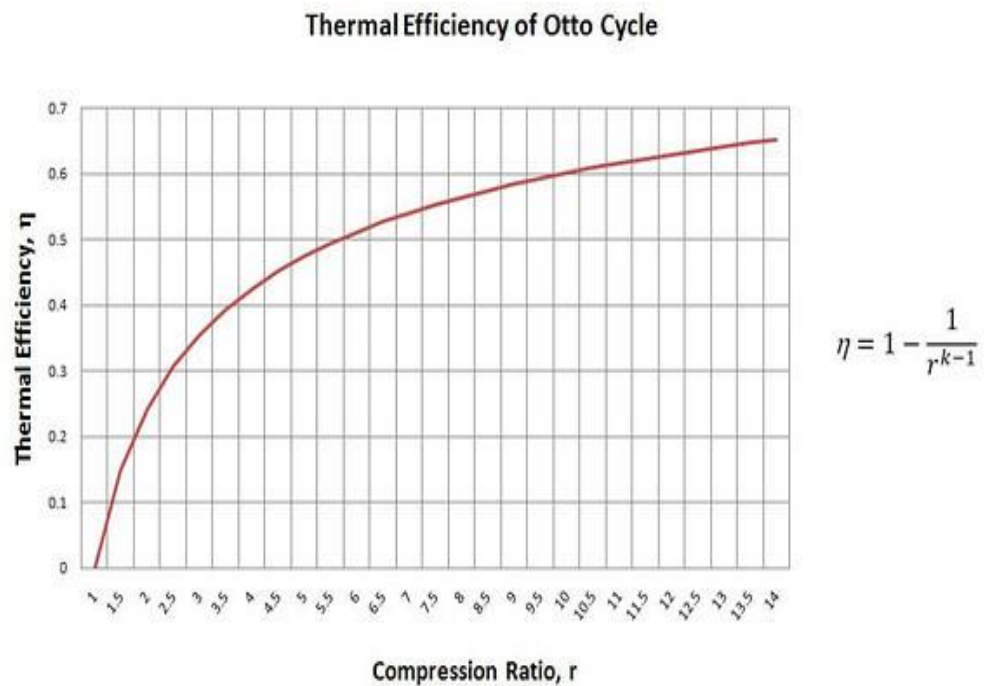


Figure 2.- Compression Ratio and Thermal Efficiency

- The compression ratio of an engine directly dictates the thermal efficiency of the engine
- The relation between compression ratio and thermal efficiency is given as

$$\eta = 1 - \frac{1}{r^n}$$

where, η is thermal efficiency and
 r is the compression ratio

- As can be seen by the equation and the graph above, the higher the compression ratio higher is the thermal efficiency of the engine.

- At lower values of compression ratio the rate of increase of thermal efficiency is very high and the rate of increase of efficiency decreases.

3.2 Compression ratio and Power

- The relation between the power output and compression ratio is similar to thermal efficiency.
- The power output of an engine increases with increase in compression ratio but only up to a certain point.
- Higher the compression ratio, cleaner the fuel burns and more is the power extracted from the fuel.
- However the peak pressure and in turn the peak temperature also increases.
- This increases the tendency of the fuel to self ignite or cause a 'knock'
- A knock occurs when another flame front, apart from the one at the spark plug is generated due to self ignition of engine.
- When these two flame fronts meet it creates large vibration in the combustion chamber.
- Thus the compression ratio can be increased only up to certain extent depending on the octane rating of the fuel

3.3 Compression Ratio and Turbocharger

- The turbocharger uses the exhaust gas circulation or the some power from the engine itself, as in case of supercharger, to increase the amount of air entering into the combustion chamber.
- The turbocharger increases the power output of the engine.
- However the compression ratio poses a limit on the turbo boost.

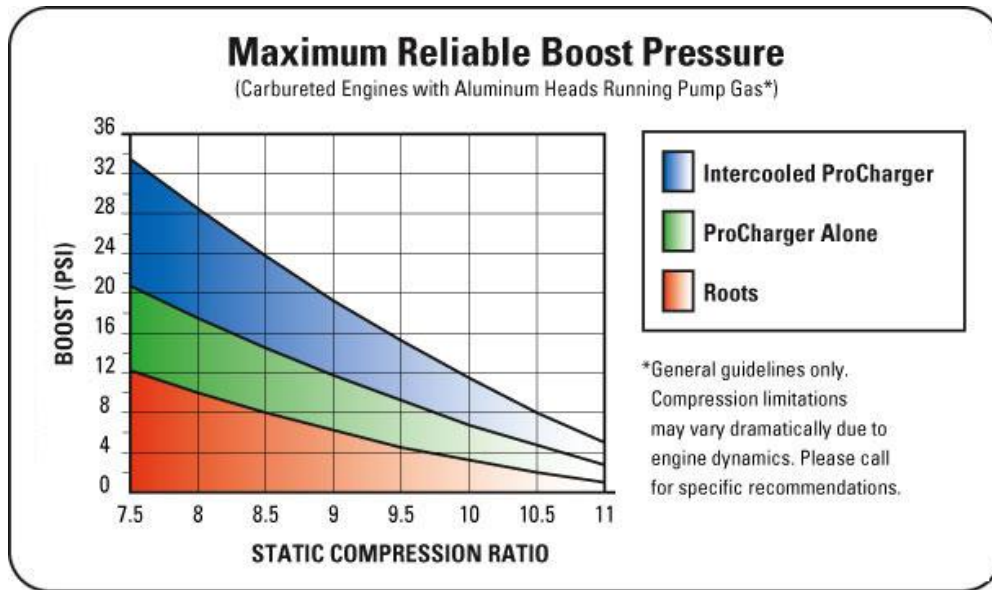


Figure 3.-Maximum Reliable boost pressure graph

- As can be seen in the figure the amount of boost pressure that can be used for high compression ratios is very less [2].
- This is due to the fact that that increase in charge entering into the engine increases the dynamic compression ratio.
- Thus to achieve higher power outputs from turbocharger, the compression ratio must be less.

3.4 Compression ratio and Brake Specific Fuel Consumption

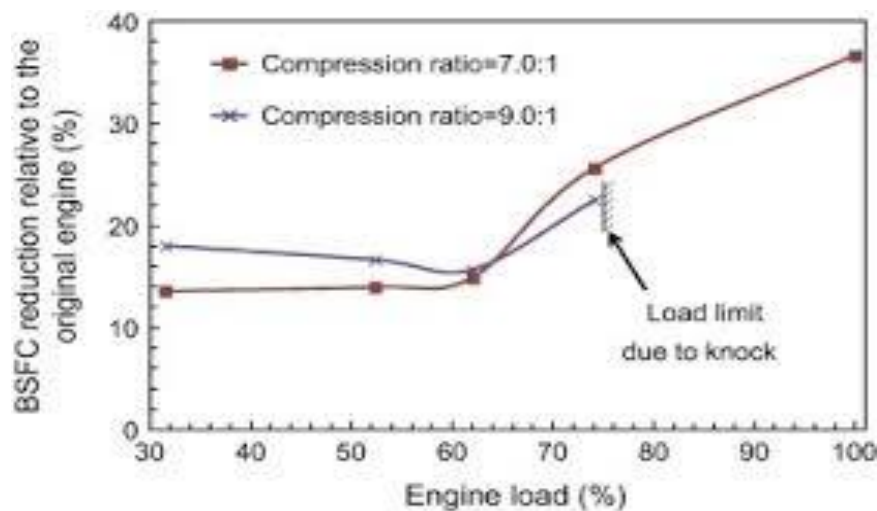


Figure 4.-Engine Load & BSFC

- It can be seen from the figure above that the brake specific fuel consumption reduces with increase in engine load.
- Thus the engine performs better at high load applications.
- However when the compression ratio is more it poses a limit on the amount of increase in the load and in turn limits brake specific fuel consumption at a higher value.
- The following table gives the strategies for reducing fuel consumption-

Fuel Consumption Reduction Strategy	Sources of effectiveness	Engine strategies
1 - Increased effective expansion ratio	Improved indicated efficiency	•High Fixed Compression Ratio (FCR) •Variable Compression Ratio
2 - Downsizing	•Pumping losses reduction •Internal heat transfer surface reduction •Flame travel distance reduction •Friction losses reduction	•High supercharging •Increased average BMEP •10:1 Compression Ratio (FCR) •From 8:1 to 16:1 Compression Ratio (VCR)
3 - Downsizing	•Pumping losses reduction •Friction losses reduction	•High supercharging •Increased average BMEP
4 - Unthrottled load control	•Pumping losses reduction	•Load control by air/fuel ratio •Load control by Late Intake Valve Closing
5 - Engine mechanical efficiency improvement	•Friction losses reduction	•Reduction of all friction sources

Figure 5.-Strategies to increase fuel efficiency

Chapter 4

What is VCR?

4.1 Overview

- Variable compression ratio offers a number of benefits for diesel and gasoline engines. While the concept has been studied for many years, the added cost and complexity have been difficult to justify for many applications. Developments that have provided a simpler mechanism through the use of variable length connecting rods appear to make this a viable option for volume production.
- The mechanisms for varying compression ratio include either a two-step system where a low or a high compression ratio can be selected or a continuously variable system where any compression ratio between a low and high value can be selected.
- Two-step systems include AVL's and FEV's variable length connecting rod. Continuously variable systems can be implemented through a variety of mechanisms including [4]:
 1. Multilink mechanism between the crankshaft and piston, Nissan and MCE-5
 2. Movable head/cylinder, SAAB [Nilsson 1995] and Enerva [Mendler 2015]
 3. Eccentric crankshaft journals, Caterpillar [Green 2002][Duffy 2006][Lawrence 2008]
 4. Movable piston crown [Brevick 1998]

4.2 Difficulty in operation

- The variable compression ratio (vcr) engines, though desirable are tough to achieve in dynamic conditions.
- There are more moving parts in the vcr engines which means more maintenance.
- Also the multilink crankshaft or articulated pistons are very expensive to manufacture.

- A very complex electronic harmonic drive system is needed to change the compression ratio whilst in operation.

4.3 Kinematic analysis of multilink crankshaft mechanism

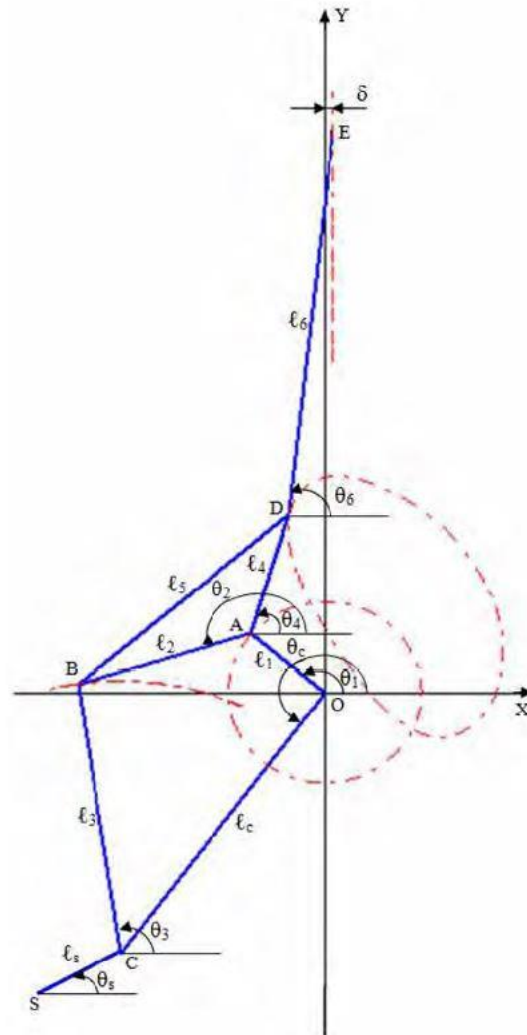


Figure 6.-Equivalent Line diagram of multilink mechanism

- A schematic of the multiple-link mechanism is shown in Figure 1. Origin O is the crankshaft axis. The crank pin A and piston pin E are connected by means of an upper link DE and a lower link ABD. The lower link is also connected, via a control link BC, to the eccentric journal C of a control shaft SC. When the crankshaft OA rotates, the lower link revolves around the crankshaft axis and pivots

around the crank pin. Meanwhile, motion of the lower link is constrained by the control link. Note the crankshaft OA, the lower link AB, the control link BC, and segment OC which can be considered a ground link, form a four-bar linkage. As a result, when the crankshaft rotates, the control link acts as a rocker and rotates through a limited range of angles around the eccentric journal of the control shaft. The connection point D of the lower and upper links traces roughly an oval shape and the piston pin E undergoes reciprocal motion [3].

- The control shaft SC is supported by the engine block. The distance between the center of the eccentric journal C and the control shaft axis S is called control shaft length in this study, and denoted by l_s . When the angle of the control shaft relative to the engine block θ_s is changed, the position of the control link will change, which subsequently tilts the lower link clockwise or counter clockwise and thereby affects the motion of the piston and shifts the TDC and BDC positions. This leads to change in both the combustion chamber volume and the swept volume. By manipulating the control shaft angle θ_s the mechanism can realize different compression ratios and displacements for various engine operating conditions.
- Note that in order for the multiple-link mechanism to operate as desired, certain constraints need to be satisfied for its geometric dimensions over the whole range of the control action θ_s . A major requirement is that the lengths of links of the four bar linkage OABC need to satisfy the Grashoff Condition, which states that the sum of the lengths of the shortest and longest links need be less than that of the remaining links. This constraint needs to be verified during design of the mechanism dimensions.
- Other constraints include the distance through which the piston must move upon actuation of lower link
- The Code attached at the back simulates the working of multilink mechanism

Chapter 5

Case Study: Infiniti Nissan VCR-Turbo engine

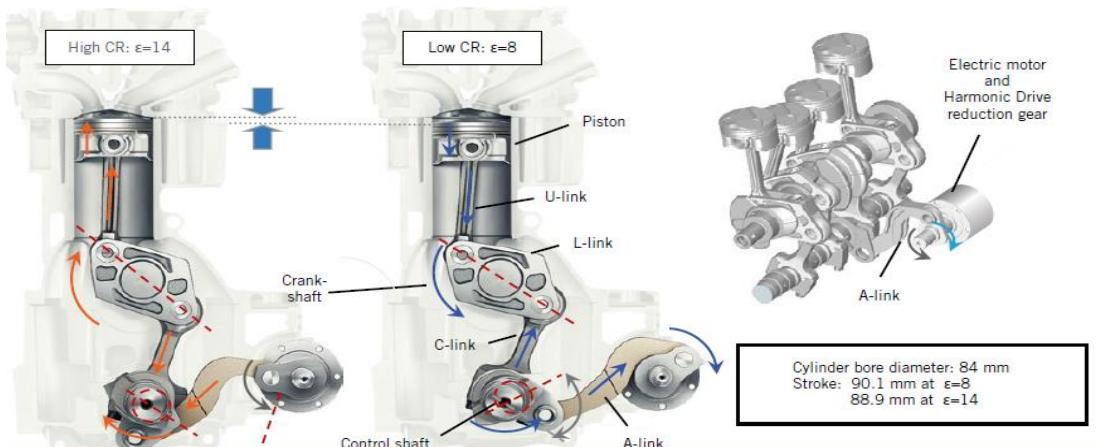
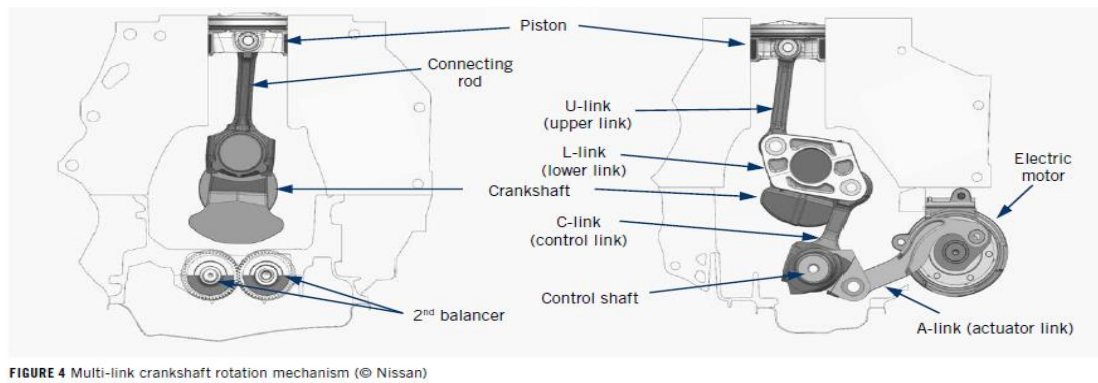
5.1 Overview

Nissan has created the world's first mass-produced VCR engine with the following improvements aimed at system compactness[1]:

- Compact L-link contributes to engine width reduction, which leads to stress reduction of L-link and NVH improvement.
- Control shaft relocation directly under the crankshaft resulted in reduction of engine width and various structure optimizations under crankshaft.
- Harmonic Drive reduction gear enabled VCR actuator downsizing and improved NVH performance.
- The VC-Turbo eliminates the traditional trade-offs imposed on conventional engines caused by having a fixed compression ratio. By changing the compression ratio according to engine operating conditions, fuel consumption, output and exhaust emissions performance are all improved to an unprecedentedly high level.

5.2 Construction of the VCR-Turbo Engine

- In this VCR engine, the conventional mechanism is replaced with Nissan's proprietary multi-link mechanism. The multi-link mechanism contains the piston and crankshaft as in the conventional mechanism, but the piston and crankshaft are linked with two links connected in series: U-link (upper link) and L-link (lower link) with the U-pin (pin connecting U-link and L-link). The motion of these two links is restrained by the C-link (control link) and control shaft, so that the included angle of these two links does not change erroneously.



- The control shaft has a main shaft supported by the engine block, and an eccentric shaft offset relative to the main shaft.
- One end of the C-link is connected to the eccentric shaft, and the another end is connected to the L-link with the C-pin (pin connecting C-link and L-link).
- When the crankshaft rotates, the L-link swings up and down while translating left and right in accordance with rotation of the crank pin, allowing the U-link and connected piston to move up and down.
- In the case of an in-line multiple cylinder engine, the piston, U-link, L-link and C-link are installed in each cylinder, but the crankshaft and control shaft are shared by all cylinders.

5.3 Operation

- When the rotation position of the control shaft changes clockwise relative to the engine block, and its eccentric shaft (this is C-link swing point) moves downward, the L-link rotates clockwise about the crank pin, the included angle at top dead centre (TDC) point between U-link and L-link increases, and U-link and piston move upwards, increasing the engine compression ratio[1].

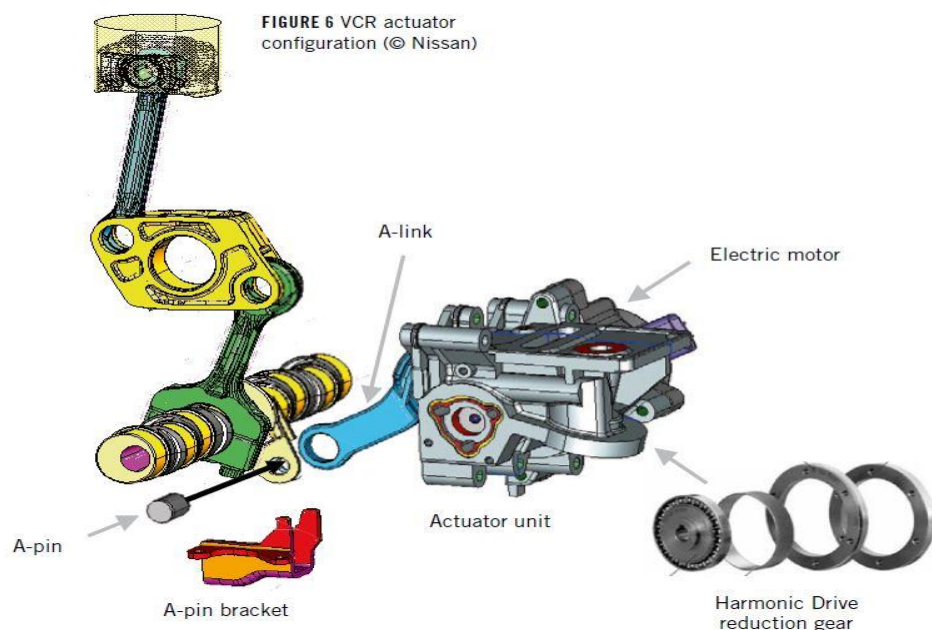


Figure 8.- Operation of VCR-Turbo

- Conversely, when the rotation position of the control shaft changes counter clockwise and C-link swing point moves upward, the L-link rotates counter clockwise about the crank pin, the included angle at TDC point between U-link and L-link decreases and the U-link and piston move downward, reducing the engine compression ratio.
- Since in an in-line multiple cylinder engine the control shaft is shared by all cylinders, the compression ratio of all cylinders can be changed simultaneously by changing rotation position of one control shaft relative to the

cylinder block. In the VCR operation of this mechanism, the TDC position of the piston changes by about 6 mm.

5.4 Salient Features

5.4.1 Reduction in side thrust during combustion stroke

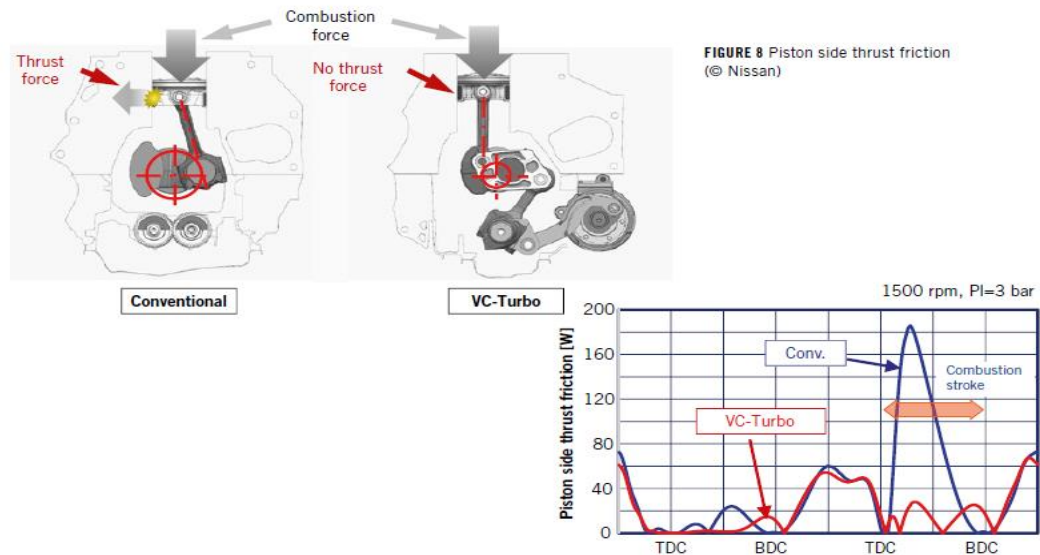


Figure 9.-Reduction in side thrust

The multi-link mechanism has another unique characteristic: The U-link is located directly under the piston on the piston down stroke (U-link: corresponds to the connecting rod of the conventional mechanism) and maintains an almost vertical position. Since the down stroke of the piston corresponds to expansion stroke during which combustion pressure is applied to the piston, when the combustion pressure is applied to the piston, the side thrust pressure to the piston caused by the inclination of the U-link located directly below the piston is significantly reduced. This greatly reduces the friction generated between side walls of the piston and the cylinder bore. As a result, this friction reduction compensates for the friction increase caused by the increase of the bearing sliding portion. Overall, a friction level that is equal

to or less than the friction level of a conventional mechanism without the secondary balancer is achieved.

5.4.2 Decrease in BSFC

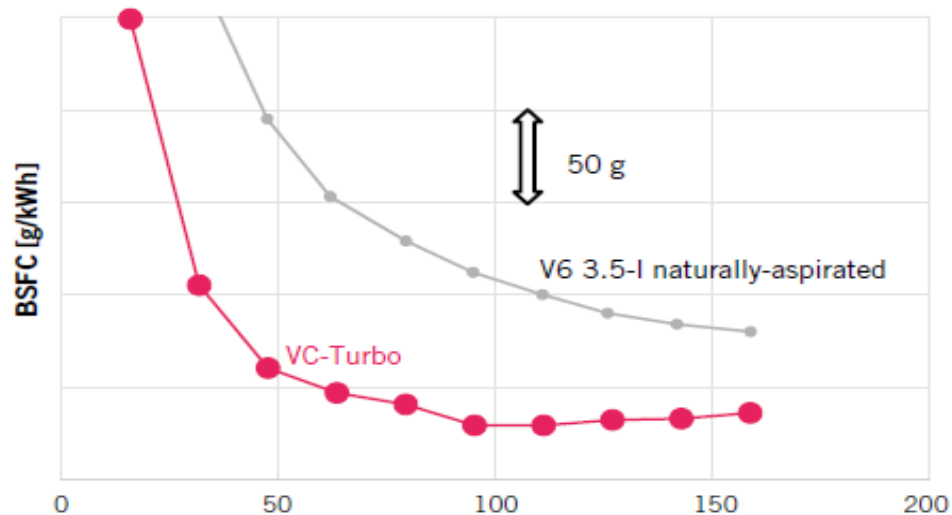


Figure 10.- Decrease in BSFC

Theoretically, increased compression ratio improves thermal efficiency in the partial load region. However, in practice these improvements in thermal efficiency are reduced by the larger cooling loss caused by increased combustion temperature. Also, enrichment is generally required to cool the chamber, reducing the improvement in thermal efficiency. Further, because at high compression ratio the piston crown surface approaches the lower surface of the cylinder head, valve clearance recesses increase the piston surface area, which becomes an additional loss factor. For this reason, it is necessary to set such compression ratio that balances these efficiency improvements and losses.

Compared to conventional engines with fixed compression ratio, the high compression ratio state of this system allows for the desired improvement in fuel consumption together with the friction reduction effect described above. Compared to the 3.5-l

V6 naturally-aspirated engine of equivalent output, the VCR improvement described above is in addition to the pumping loss and friction reduction benefits from downsizing.

5.4.3 Contribution to output and torque improvement

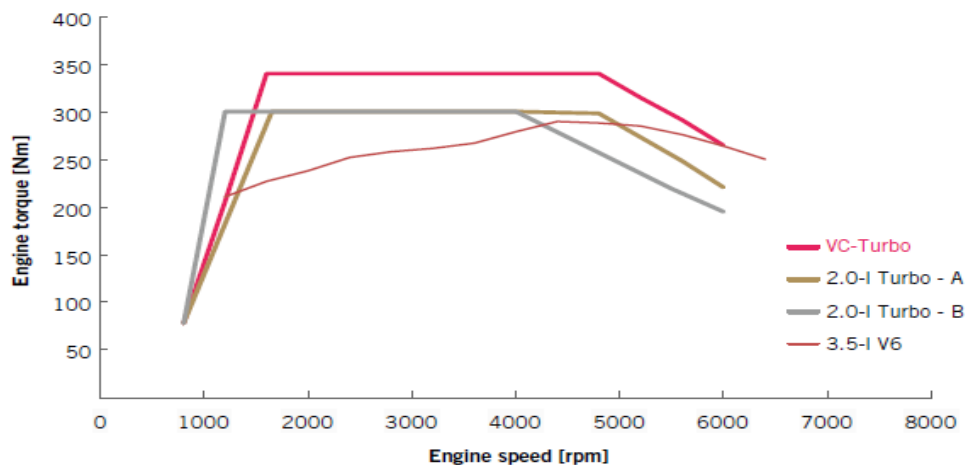


FIGURE 10 Torque curve comparison (© Nissan)

Figure 11.-Engine speed Vs Engine Torque (VCR-Turbo)

In a conventional engine at high compression ratio, boosting is limited under high load operation in order to prevent knock. Ignition retard and fuel enrichment, which are methods used to prevent knock in conventional engines, reduce thermal efficiency, reduce combustion stability and increase exhaust temperature. Moreover, since cylinder pressure becomes higher, durability may be affected due to permissible cylinder pressure and main structure limits. In this VCR engine, setting a lower compression ratio in the high load range improved torque by 10 % or more, and it is possible to achieve torque levels surpassing that of the 3.5-l V6 naturally-aspirated engine.

Chapter 6

SUMMARY

Hence the important benefits of the VCR engine can be summarized as follows [4]: —

- Optimum combustion efficiency in the whole load and speed range.
- Low fuel consumption and low exhaust emissions.
- The VCR provides better control over pollutant generation and after-treatment than a conventional fixed compression ratio (FCR) engine, also extends the life expectancy of a three way catalytic converter.
- As the geometrical volumetric ratio is under control on VCR engines, the engine always operates below the knock limit, whatever the load.
- The VCR engine provides excellent fuel flexibility, since the compression ratio can be varied and adjusted to suit the properties of the fuel, and therefore the engine will always run at the compression ratio best suited to the fuel being used for bi-fuel (compressed natural gas (CNG)/gasoline) power-trains, the realization of VCR is of specific interest.
- High fuel flexibility, with optimal combustion efficiency.
- Very smooth idle and full load accelerations are achieved.
- It provides better indicated thermal efficiency than that of FCR engines.
- It allows for a significant idle speed reduction because of reduced misfiring and cyclic irregularities, resulting in low vibration levels.
- Reduction in low-frequency noise because of constant peak pressures.
- Smoother combustion because the rate of heat release is the same (short) both at low and high compression ratios.
- Cold starting emissions can be reduced greatly by early catalyst warm-up in the catalytic converter.

- Improvement in the low end torque of a petrol engine without the risk of detonation.
- Potential technology for future high-boosting super lean burn engines.
- Low CO₂ emissions by down-sizing for the same power output.
- Good idling performance at low ambient temperatures.
- Constant frictional losses owing to almost constant peak pressures.

Matlab Code

```

l1=42; l2=115; l3=140; l4=58; l5=155; l6=150; ls=30; lc=50;
Xo=0; Yo=0; Xs=-110; Ys=-120;
delta=-4;
axis([-4*l1 4*l1 -4*l1 40*l1+l6]);
X_Piston=[-7 16 16 -7 -7];
Y_Piston=[l6-20 l6-20 l6+20 l6+20 l6-20];
g =
plot(X_Piston,Y_Piston,'r','LineWidth',2,'XDataSource','X_Piston','
YDataSource','Y_Piston');
hold on
theta_1=0:0.01:2*pi;
theta_s=pi-36*pi/180;
Xa=l1.*cos(theta_1);
Ya=l1.*sin(theta_1);
i = plot(Xa,Ya,'LineWidth',2)
hold on
grid on
Xc=Xs+ls*cos(theta_s);
Yc=Ys+ls*sin(theta_s);
Xcs=[Xs Xc];
Ycs=[Ys Yc];
h=plot(Xcs,Ycs,'LineWidth',2,'XDataSource','Xcs','YDataSource','Y
cs');
Xz=l1*cos(theta_1)-Xs-ls*cos(theta_s);
Yz=l1*sin(theta_1)-Ys-ls*sin(theta_s);
theta_3=atan(Yz/Xz)+acos(((l3).^2-
(l2).^2+Xz.^2+Yz.^2)/(2*l3*sqrt(Xz.^2+Yz.^2)));
theta_2=3.142-asin((l3*sin(theta_3)-Yz)/l2);
Xb=Xc+l3*cos(theta_3);
Yb=Yc+l3*sin(theta_3);
theta_4=theta_2-acos((l2.^2+l4.^2-l5.^2)/(2*l2*l4));
theta_D=asin(l2*sin(theta_2-theta_4)/l5);
theta_B=asin(l4*sin(theta_2-theta_4)/l5);
Xd=Xa+l4*cos(theta_4);
Yd=Ya+l4*sin(theta_4);
theta_6=acos((delta-Xd)/l6);

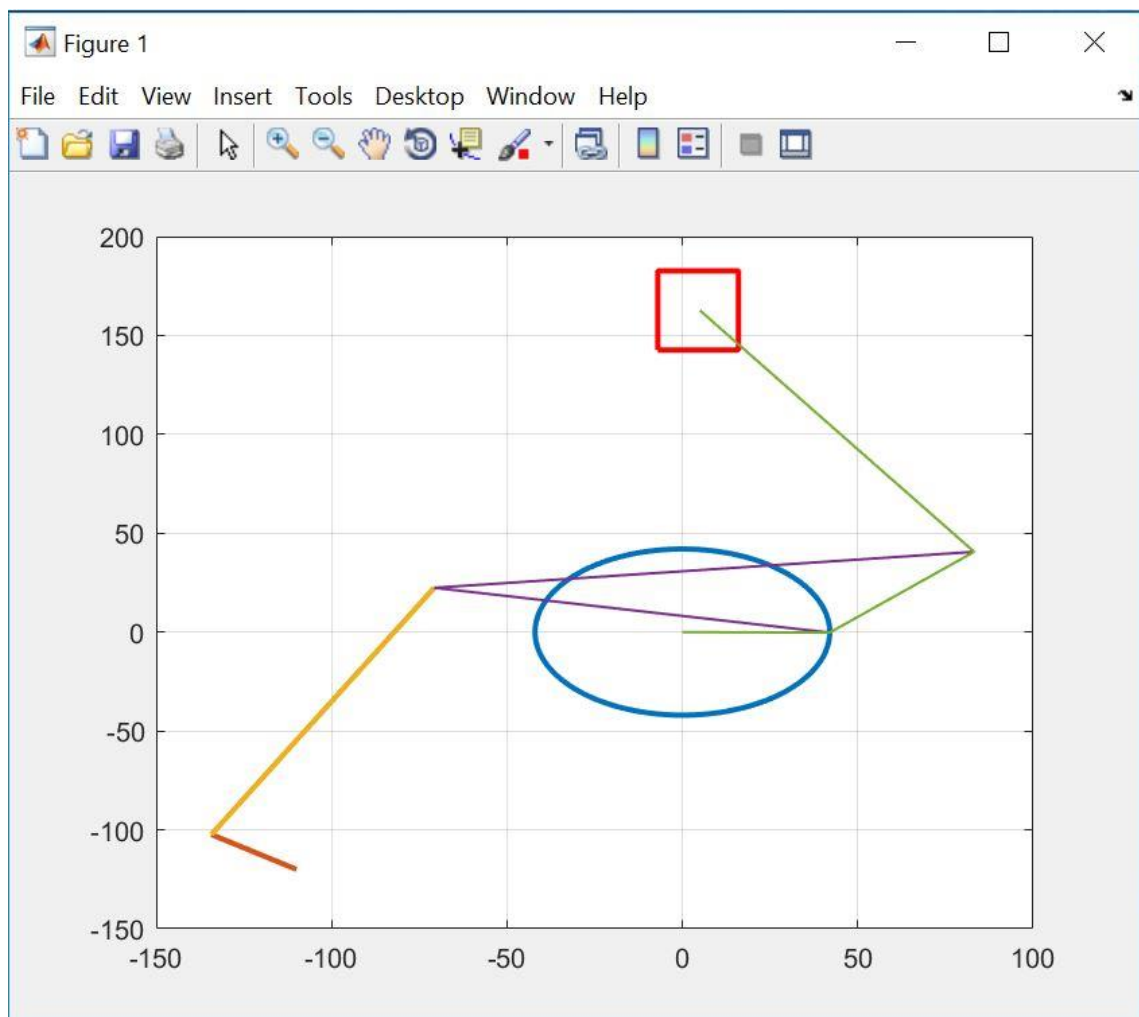
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```

Xe=ones(1,629)*5;
Ye=Yd+l6*sin(theta_6);
Xcb=[Xc Xb];
Ycb=[Yc Yb];
Xabd=[Xa Xb Xd Xa];
Yabd=[Ya Yb Yd Ya];
Xoade=[Xo Xa Xd Xe];
Yoade=[Yo Ya Yd Ye];
f=plot(Xcb,Ycb,'LineWidth',2,'XDataSource','Xcb','YDataSource','Y
cb');
r=plot(Xabd,Yabd,'LineWidth',1,'XDataSource','Xabd','YDataSource
','Yabd');
p=plot(Xoade,Yoade,'LineWidth',1,'XDataSource','Xoade','YDataSo
urce','Yoade');
hold on
for theta_1=0:0.08:4*pi
%theta_1=pi/2;
Xa=l1.*cos(theta_1);
Ya=l1.*sin(theta_1);
Xz=l1*cos(theta_1)-Xs-ls*cos(theta_s);
Yz=l1*sin(theta_1)-Ys-ls*sin(theta_s);
theta_3=atan(Yz/Xz)+acos(((l3).^2-
(l2).^2+Xz.^2+Yz.^2)/(2*l3*sqrt(Xz.^2+Yz.^2)));
theta_2=3.142-asin((l3*sin(theta_3)-Yz)/l2);
Xb=Xc+l3*cos(theta_3);
Yb=Yc+l3*sin(theta_3);
theta_4=theta_2-acos((l2.^2+l4.^2-l5.^2)/(2*l2*l4));
theta_D=asin(l2*sin(theta_2-theta_4)/l5);
theta_B=asin(l4*sin(theta_2-theta_4)/l5);
Xd=Xa+l4*cos(theta_4);
Yd=Ya+l4*sin(theta_4);
theta_6=acos((delta-Xd)/l6);
Xe=5;
Ye=Yd+l6*sin(theta_6);Xcb=[Xc Xb];
Ycb=[Yc Yb];
Xabd=[Xa Xb Xd Xa];
Yabd=[Ya Yb Yd Ya];
Xoade=[Xo Xa Xd Xe];
Yoade=[Yo Ya Yd Ye];

```

```
Y_Piston=[Ye-20 Ye-20 Ye+20 Ye+20 Ye-20];  
refreshdata(i,'caller')  
refreshdata(f,'caller')  
refreshdata(r,'caller')  
refreshdata(p,'caller')  
refreshdata(g,'caller')  
drawnow  
pause(.1)  
end
```



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